Elevator Angle Position Indicator – General data

1. General

A potentiometer is mounted on the port side elevator hinge bracket assembly at station WL 58.875 and sends a signal of the elevator angle position through the DAU and onto the Flight Data Recorder (if installed).

The potentiometer can be accessed by removing the cover panel (Mod 6/2417). The panel is secured to the aircraft by screws.

2. Removal/Installation

Equipment and Materials:

- Corrosion preventative compound MIL–C–16173 Grade 2 or equivalent

A. Remove the Elevator Angle Position Potentiometer

1) Ensure the electrical system is OFF (refer to 24-00-00, Electrical Power – General Data).

2) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

3) Remove the cover panel fasteners and remove the panel (Mod 6/2417).

4) Disconnect the electrical connection at the terminal block.

5) Remove the cotter pin, nut, (2) washers and bolt from the push rod end at the elevator hinge bracket.

6) Remove the two bolts which hold the potentiometer mounting bracket to the structure.

7) Remove the potentiometer and mounting bracket assembly.

8) Loosen the nut and screw that hold the lever assembly to the potentiometer shaft mounted in mounting bracket.

9) Remove the safety wire and the (3) screws and clamps which secure the potentiometer onto the mounting bracket.

10) Remove the potentiometer.

B. Install the Elevator Angle Position Potentiometer

Note Coat all bolt shanks with corrosion preventative compound MIL–C–16173 Grade 2 or equivalent before installation.

1) Ensure the electrical system is OFF (refer to 24-00-00, Electrical Power – General Data).

2) Ensure the FLIGHT CONTROLS INDICATION (A14) circuit breaker is pulled and fitted with safety clip.

3) Position the potentiometer on to the mounting bracket. Secure with the (3) clamps and screws. Safety wire screws.

4) Locate the lever assembly on potentiometer shaft leaving a gap of 0.02 to 0.05 inch between the lever assembly and the potentiometer housing to minimize the push rod assembly misalignment.

5) Secure with a screw, washer and nut.

6) Install bolt through the push rod assembly at lever end, install washer, nut and cotter pin.

7) Position the mounting bracket and secure in place with bolt, and washers in two places.

8) Install nut and cotter pin bolt through the elevator hinge end. Install washer, nut and cotter pin.

9) Adjust the elevator angle position potentiometer. (Refer to Para 3.A).

10) Replace the cover panel (Mod 6/2417) and re-install fasteners. Do not overtighten.

11) Reconnect the electrical connection at terminal block.

12) Remove the safety clip and reset FLIGHT CONTROLS INDICATION (A14) circuit breaker.

13) Do an Elevator Position Voltage Check per Para 3.B.

3. Adjustment/Test

A. Adjustment Elevator Angle Position Potentiometer

Equipment and Material:

- DC Voltmeter

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Measure the Ground Power Voltage (V_{GP}) at the external power relay (or other suitable location).

3) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

4) Move the elevator to the neutral position and lock in place.

5) Disconnect terminal 2 of the potentiometer electrical connection at the terminal block (TB5) and connect voltmeter positive to the lead from terminal 2 while attaching voltmeter negative to aircraft ground.

CAUTION **ENSURE THE TERMINAL 2 LEAD DOES NOT COME INTO CONTACT WITH AN ELECTRICAL GROUND.**

Note There are a set of drop-down resistors at the DAU. If reading the voltage at the DAU or on the ASCB bus, this voltage is reduced to 0.5 times of the values listed in this section.

6) Loosen screw and nut on the lever at the potentiometer shaft. Remove lever from shaft.

7) Reset the FLIGHT CONTROLS INDICATION (A14) circuit breaker.

8) Rotate potentiometer shaft until the measured DC Voltage is precisely 50% of total aircraft voltage V_{GP} (approximately 14 volts).

9) Reconnect pushrod and lever assembly to potentiometer shaft.

10) Tighten screw and nut on lever leaving a gap of 0.02 to 0.05 of an inch between lever assembly and potentiometer housing.

Note Do not rotate potentiometer shaft during this process.

11) Remove elevator control lock.

12) With voltmeter connected, function test elevator control position per the following:

a) Deflect elevator to the full up position (Reference 25 degrees) and verify measured voltage is 91.70% of V_{GP}.

b) Deflect elevator to the full down position (Reference 16 degrees) and verify measured voltage is 23.30% of V_{GP}.

13) Pull FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip..

14) Disconnect voltmeter from potentiometer.

15) Reconnect the potentiometer electrical connections at the terminal block.

16) Remove safety clip and reset FLIGHT CONTROLS INDICATION (A14) circuit breaker.

17) Deflect the elevator full up and ensure DAU Maintenance Status CAS message does not activate.

18) Deflect the elevator full down and ensure DAU Maintenance Status CAS message does not activate.

19) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

Note For Flight Data Recorder input checks, refer to 31-31-00, Flight Data Recorder (FDR) – General Data.

For Data Acquisition Unit input checks, refer to 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data.

B. Elevator Position Voltage Check

Equipment and Material:

- DC Voltmeter

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Measure the Ground Power Voltage (V_{GP}) at the external power relay (or other suitable location).

3) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

4) Move the elevator to the neutral position and lock in place.

5) Disconnect terminal 2 of the potentiometer electrical connection at the terminal block (TB5) and connect voltmeter positive to the lead from terminal 2 while attaching voltmeter negative to aircraft ground.

CAUTION ENSURE THE TERMINAL 2 LEAD DOES NOT COME INTO CONTACT WITH AN ELECTRICAL GROUND.

Note There are a set of drop-down resistors at the DAU. If reading the voltage at the DAU or on the ASCB bus, this voltage is reduced to 0.5 times of the values listed in this section.

6) Reset the FLIGHT CONTROLS INDICATION (A14) circuit breaker and confirm that DC Voltage is precisely 50% of total aircraft voltage V_{GP} (approximately 14 volts).

7) If DC Voltage is not 50% of V_{GP}, do adjustment steps per Para 3.A.

8) Remove elevator control lock.

9) With voltmeter connected, function test elevator control position per the following:

a) Deflect elevator to the full up position (Reference 25 degrees) and verify measured voltage is 91.70% of V_{GP}.

b) Deflect elevator to the full down position (Reference 16 degrees) and verify measured voltage is 23.30% of V_{GP}.

10) If DC Voltage does not correspond, do adjustment steps per Para 3.A.

11) Pull FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit with safety clip.

12) Disconnect voltmeter from potentiometer.

13) Reconnect the potentiometer electrical connections at the terminal block.

14) Remove safety clip and reset FLIGHT CONTROLS INDICATION (A14) circuit breaker.

15) Deflect the elevator full up and ensure DAU Maintenance Status CAS message does not activate.

16) Deflect the elevator full down and ensure DAU Maintenance Status CAS message does not activate.

Note For Flight Data Recorder input checks, refer to 31-31-00, Flight Data Recorder (FDR) – General Data.

For Data Acquisition Unit input checks, refer to 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data.

C. Alignment of Mechanical Pitch Trim Pointer and EFIS Pitch Trim Display Operational Test

1) Rotate the pitch trim handwheel until the pointer indicates the aircraft full nose up position while monitoring the Electronic Flight Instrument System (EFIS) Pitch Trim Display.

2) Ensure that the EFIS display indicates full pitch up (25 degrees ± 1 degree) at the same time the mechanical pointer indicates full nose up. Visually confirm the Elevator Trim Tab is fully deflected down.

3) Rotate the pitch trim handwheel until the pointer indicates the full nose down position while monitoring the EFIS Pitch Trim Display.

4) Ensure that the EFIS display indicates full pitch down (20 degrees ± 1 degree) at the same time as the mechanical pointer indicates full nose down. Visually confirm the Elevator Trim Tab is fully deflected up (20 degrees ± 1 degree).

5) Adjust the pitch trim handwheel until the pointer indicates the neutral position while monitoring the EFIS Pitch Trim Display.

6) Ensure that the EFIS display indicates the neutral position (0 degrees) at the same time the mechanical pointer indicates the neutral pitch trim. Confirm the Elevator Trim Tab is at the neutral position.

7) If adjustment of the EFIS Pitch Trim Display Potentiometer is required refer to 27-33-21, Elevator Trim Position Transmitter – General Data Para 3.A. If adjustment of the mechanical pitch trim display is required, reference the mechanical trim system rigging, in accordance with 27-33-00, Elevator Trim Tab System – General Data Para 5.A.

Note The mechanical pitch trim display is the primary trim indication system.